R: Introduction and review

EC421, Set 01(R)

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Econometrics

Recall: Applied econometrics, data science, analytics require:

1. Intuition for the **theory** behind statistics/econometrics (assumptions, results, strengths, weaknesses).
2. Practical knowledge of how to **apply theoretical methods** to data.
3. Efficient methods for **working with data** (cleaning, aggregating, joining, visualizing).

This course aims to deepen your knowledge in each of these three areas.

- 1: As before.
- 2-3: **R**

R

R

What is R?

To quote the R project website:

> R is a free software environment for statistical computing and graphics. It compiles and runs on a wide variety of UNIX platforms, Windows and MacOS.

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What does that mean?

- R was created for the statistical and graphical work required by econometrics.
- R has a vibrant, thriving online community. (stack overflow)
- Plus it's **free** and **open source**.

R

Why are we using R?

1. R is **free** and **open source** saving both you and the university 💰💵💰.
2. *Related:* Outside of a small group of economists, private- and public-sector **employers favor R** over `Stata` and most competing softwares.
3. R is very **flexible and powerful** adaptable to nearly any task, *e.g.*, metrics, spatial data analysis, machine learning, web scraping, data cleaning, website building, teaching (these slides).

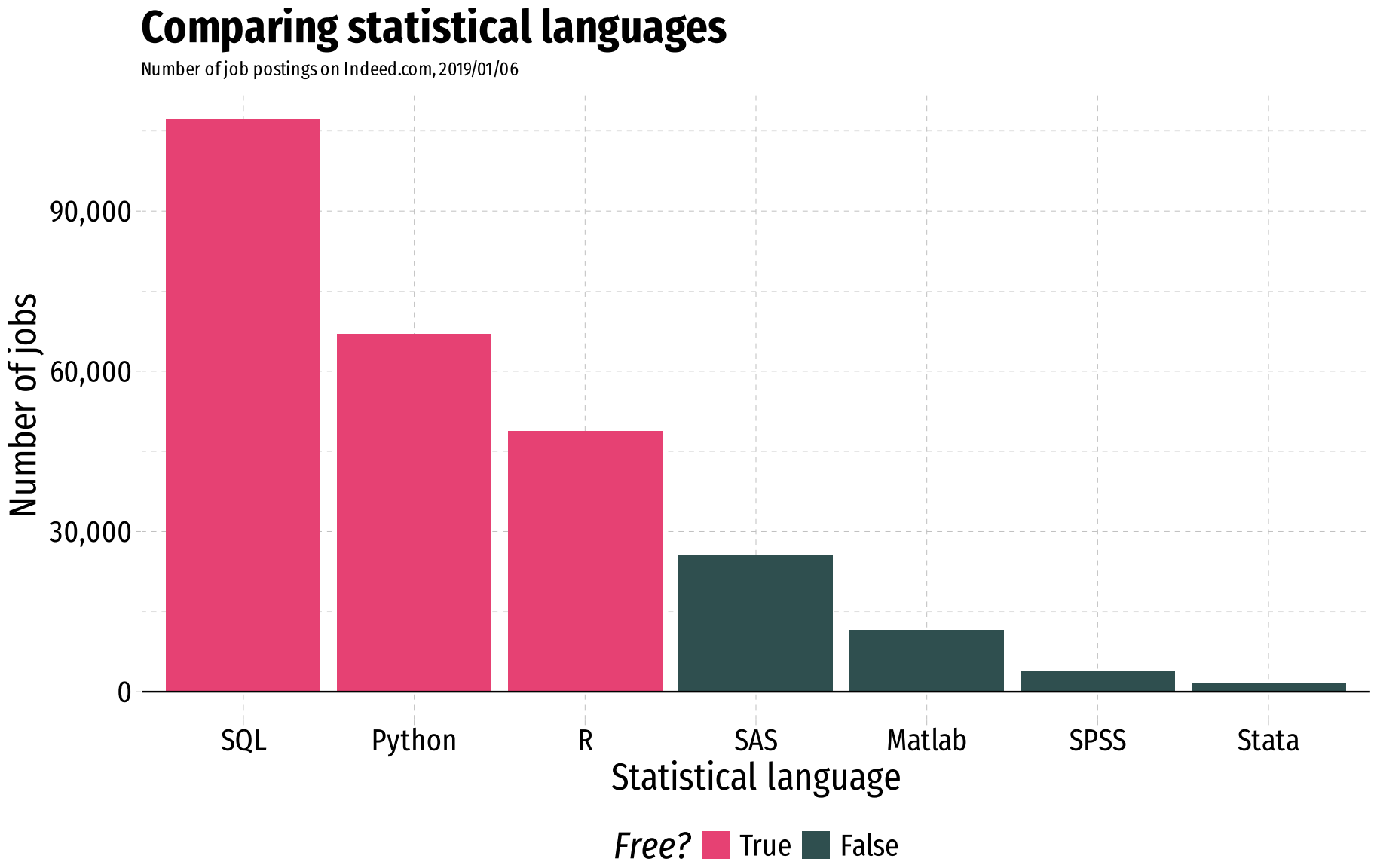
R

Why are we using R?

4. *Related:* R imposes **no limitations** on your amount of observations, variables, memory, or processing power. (I'm looking at **you**, `Stata`.)
5. If you put in the work,[†] you will come away with a **valuable and marketable** tool.
6. I 💖 **R**
7. R is a nice gateway to (and plays well with) other programming languages (*e.g.*, Python, SQL, C++, JavaScript).

[†] Learning R definitely requires time and effort.

R


Comparing statistical languages
Number of job postings on Indeed.com, 2019/01/06
Number of jobs
90,000
60,000
30,000
0
SQL
Python
R
SAS
Matlab
SPSS
Stata
Statistical language
Free?
True
False

R + Examples

R + Regression

```r
# Fit a linear regression
fit = lm(dist ~ 1 + speed, data = cars)
# Show the results
coef(summary(fit))
```

```
              Estimate Std. Error   t value     Pr(>|t|)
(Intercept) -17.579095  6.7584402 -2.601058 1.231882e-02
speed         3.932409  0.4155128  9.463990 1.489836e-12
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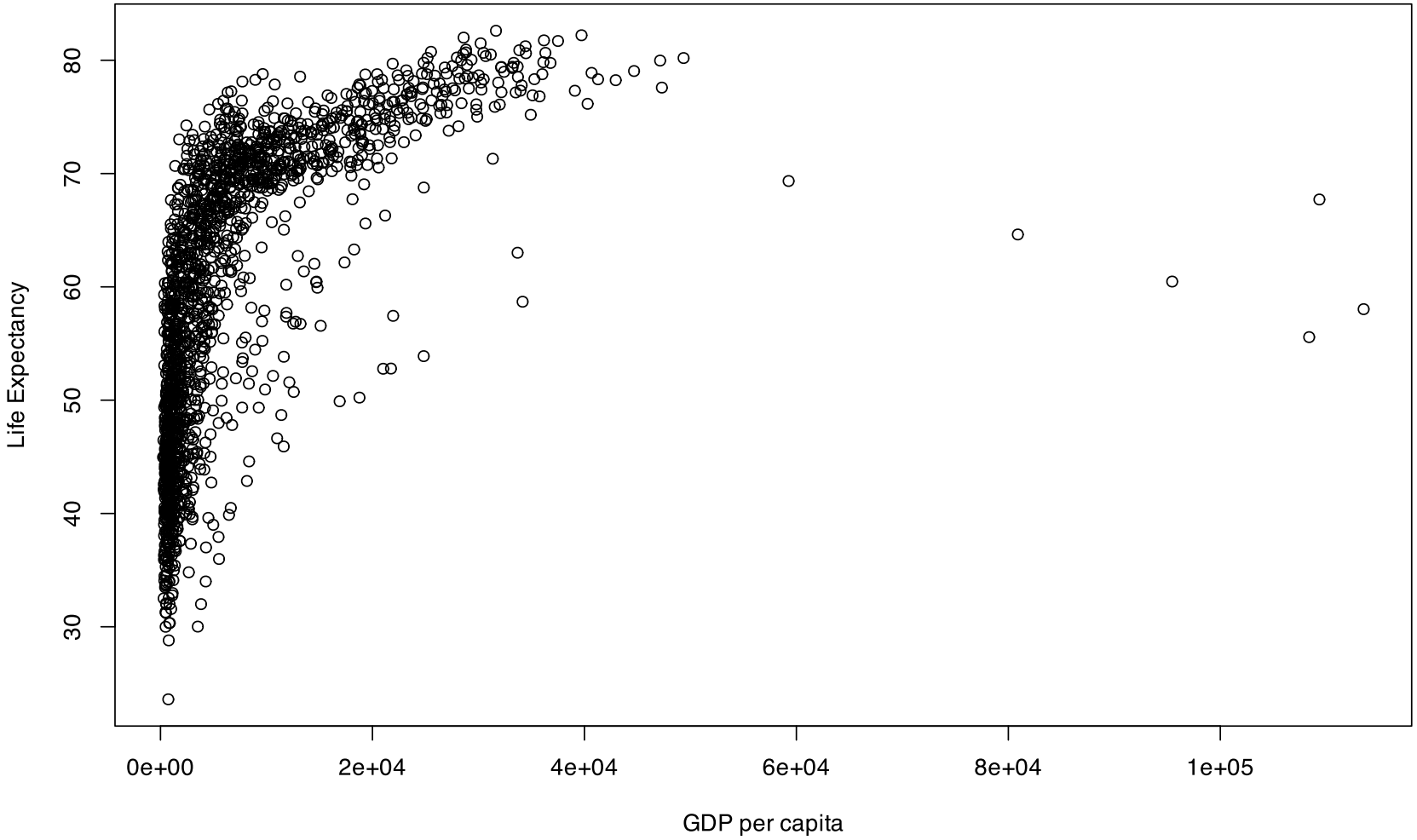
```r
# 'tidy' the results
library(broom)
tidy(fit)
```

```
# A tibble: 2 × 5
  term        estimate std.error statistic  p.value
  <chr>          <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)   -17.6      6.76      -2.60 1.23e- 2
2 speed           3.93     0.416      9.46 1.49e-12
```

R + Plotting (w/ `plot`)

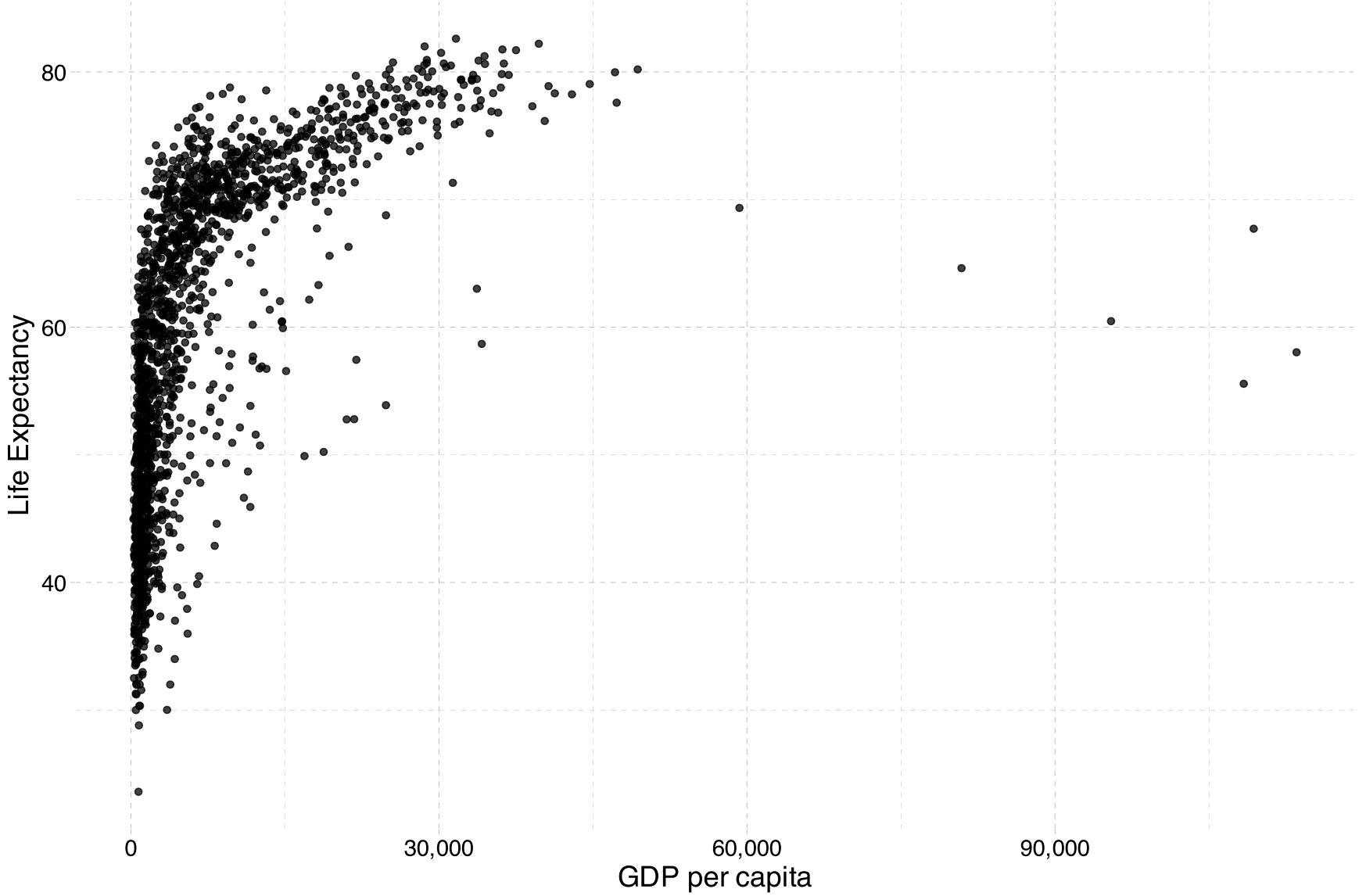


R + Plotting (w/ plot)

```r
# Load packages with dataset
library(gapminder)

# Create simple plot
plot(
  x = gapminder$gdpPercap,
  y = gapminder$lifeExp,
  xlab = 'GDP per capita',
  ylab = 'Life Expectancy'
)
```

R + Plotting (w/ ggplot2)

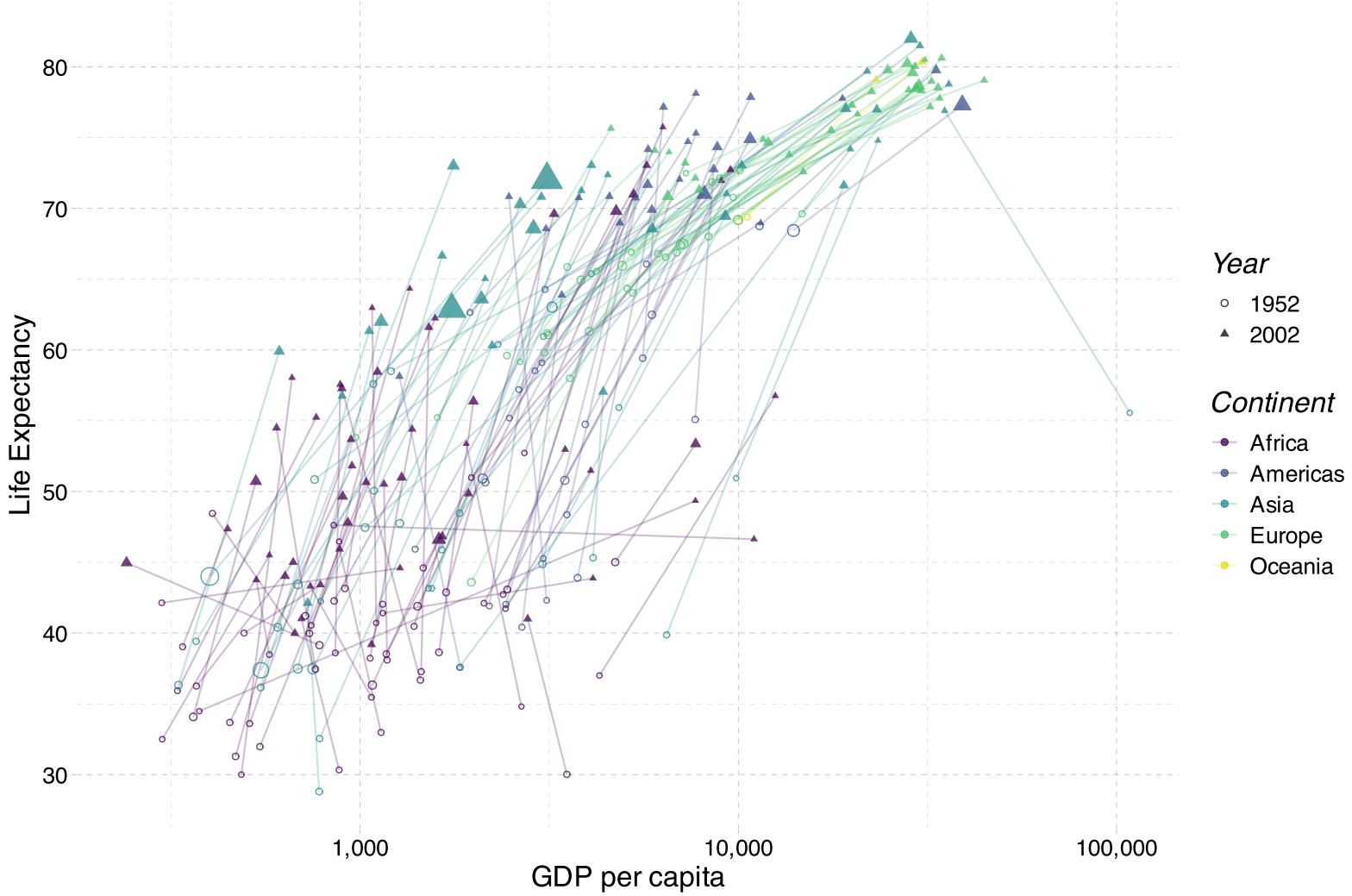


R + Plotting (w/ ggplot2)

```r
# Load packages
library(gapminder)
library(dplyr)

# Plot with ggplot2
ggplot(data = gapminder, aes(x = gdpPercap, y = lifeExp)) +
  geom_point(alpha = 0.75) +
  scale_x_continuous('GDP per capita', label = scales::comma) +
  ylab('Life Expectancy') +
  theme_pander(base_size = 16)
```

R + More plotting (w/ ggplot2)

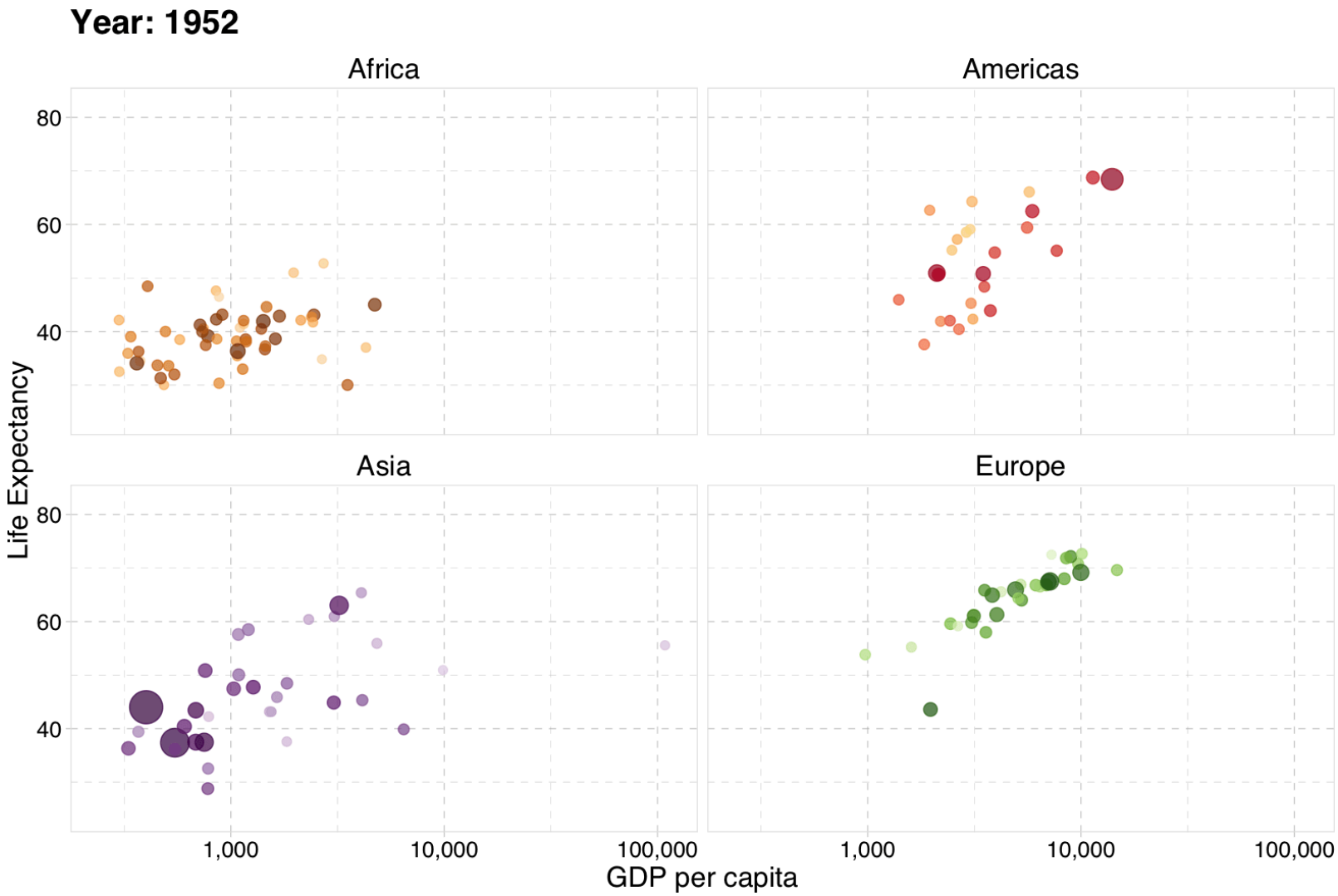


Notice we've moved to *log-10 scale* on the x-axis.

R + More plotting (w/ ggplot2)

```r
# Plot with ggplot2
ggplot(
  data = filter(gapminder, year %in% c(1952, 2002)),
  aes(x = gdpPercap, y = lifeExp, color = continent, group = country)
) +
  geom_path(alpha = 0.25) +
  geom_point(aes(shape = as.character(year), size = pop), alpha = 0.75) +
  scale_x_log10('GDP per capita', label = scales::comma) +
  ylab('Life Expectancy') +
  scale_shape_manual('Year', values = c(1, 17)) +
  scale_color_viridis('Continent', discrete = TRUE, end = 0.95) +
  guides(size = FALSE) +
  theme_pander(base_size = 16)
```

R + Animated plots (w/ gganimate)

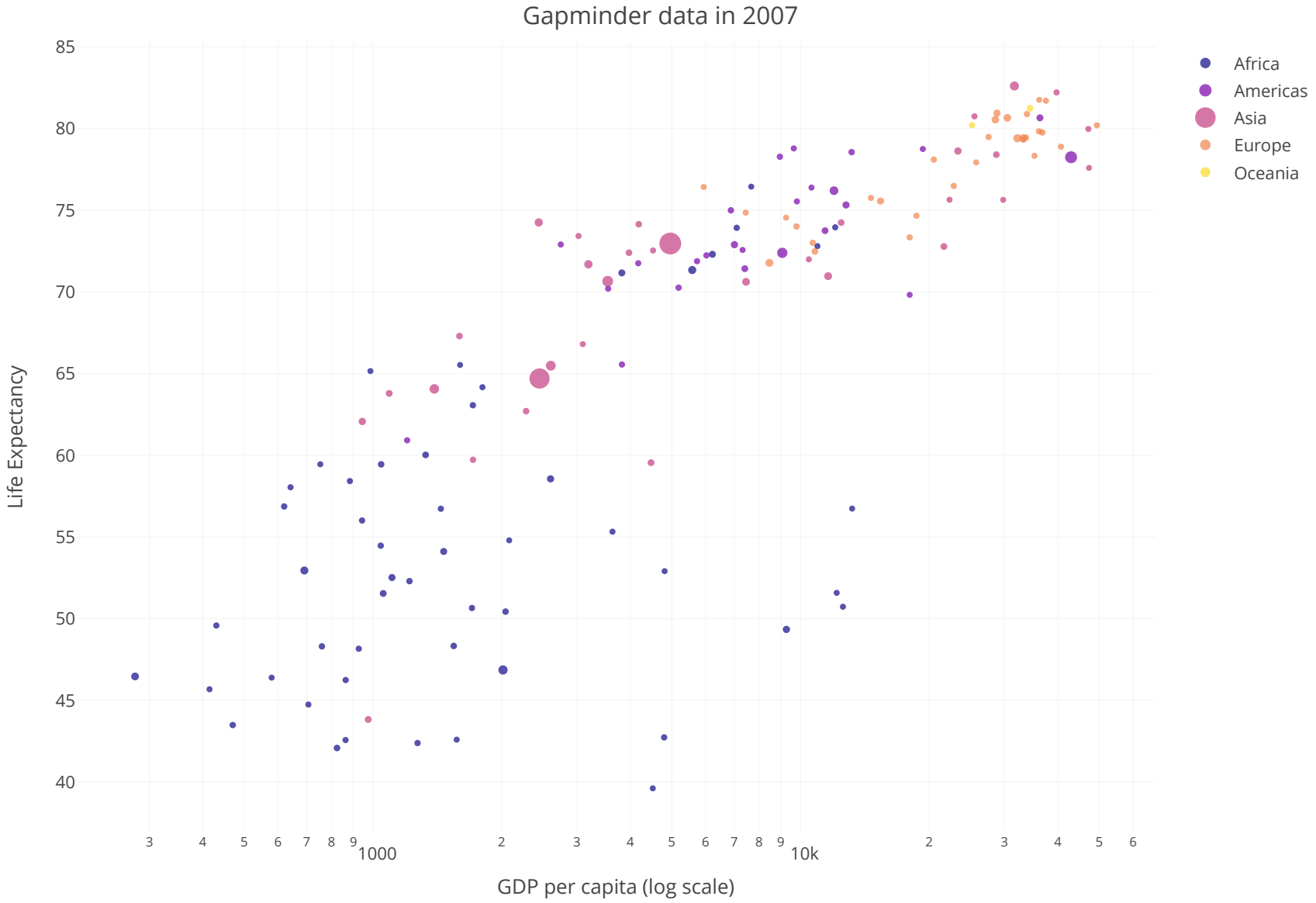


Gapminder

R + Animated plots (w/ gganimate)

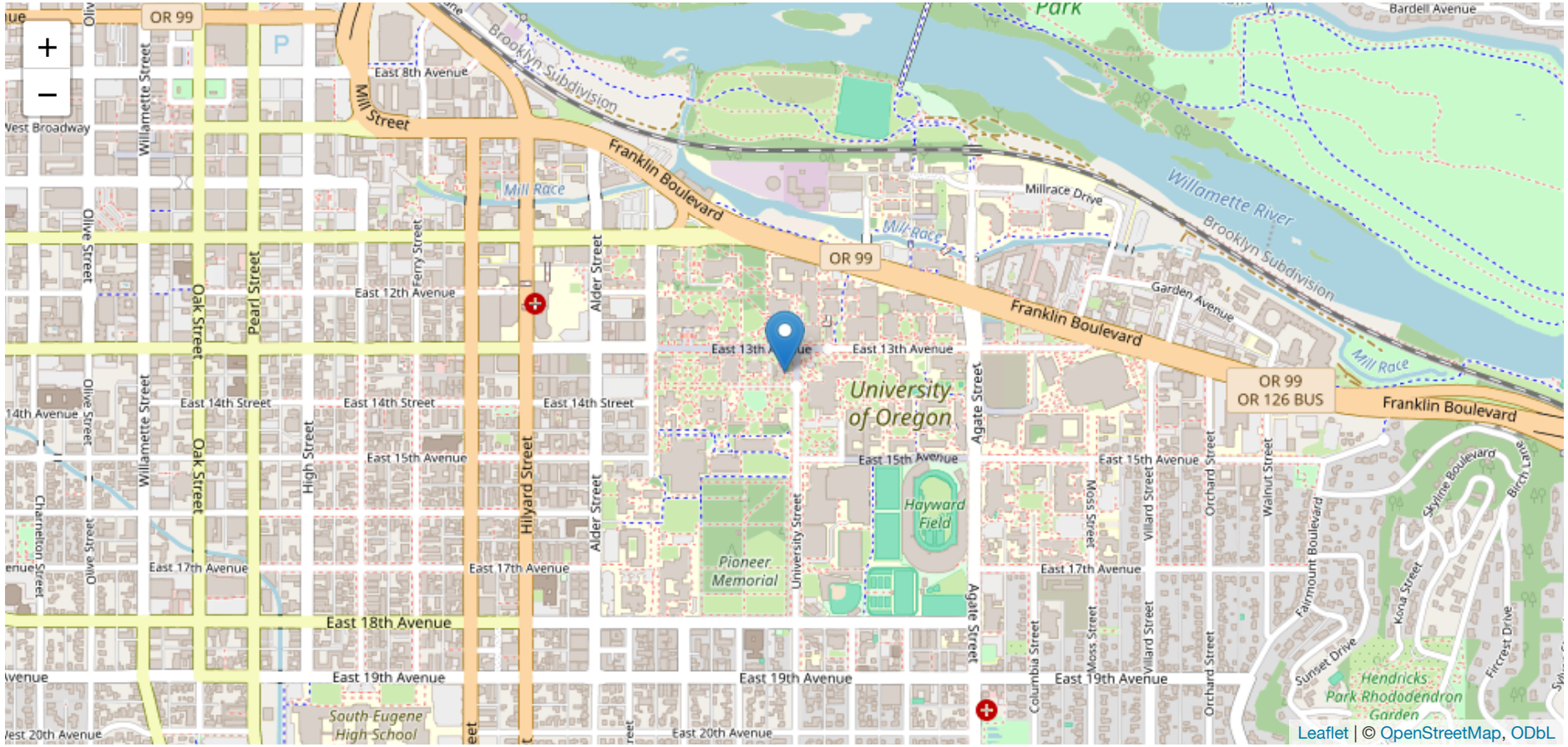
```r
# Load gganimate
library(gganimate)
# Create the plot
ggplot(
  data = gapminder %>% filter(continent != 'Oceania'),
  aes(gdpPercap, lifeExp, size = pop, color = country)
) +
  geom_point(alpha = 0.7, show.legend = FALSE) +
  scale_colour_manual(values = country_colors) +
  scale_size(range = c(2, 12)) +
  scale_x_log10('GDP per capita', label = scales::comma) +
  facet_wrap(~continent) +
  theme_pander(base_size = 16) +
  theme(panel.border = element_rect(color = 'grey90', fill = NA)) +
  labs(title = 'Year: {frame_time}') +
```

R + Interactive plots (w/ `plotly`)



```r
plot_ly(
  data = gapminder %>% filter(year == 2007),
  x = ~gdpPercap,
  y = ~lifeExp,
  type = 'scatter',
  mode = 'markers',
  size = ~pop,
  color = ~continent,
  colors = viridis::plasma(n = 5, end = .93),
  text = ~paste(
    'Country: ', country,
    '<br>GDP per capita:', scales::dollar(gdpPercap, 1),
    '<br>Life Expectancy:', scales::comma(lifeExp, 1),
    '<br>Population:', scales::comma(pop)
  )
```

R + Maps


OR 99
Franklin Boulevard
Willamette River
Brooklyn Subdivision
Mill Race
Millrace Drive
Garden Avenue
OR 99
OR 126 BUS
Franklin Boulevard
University of Oregon
Hayward Field
Pioneer Memorial
Hendricks Park Rhododendron Garden
South Eugene High School
East 8th Avenue
East 12th Avenue
East 13th Avenue
East 14th Street
East 15th Avenue
East 17th Avenue
East 18th Avenue
East 19th Avenue
East 20th Avenue
Mill Street
Willamette Street
Olive Street
Oak Street
Pearl Street
Ferry Street
Alder Street
High Street
Hilyard Street
University Street
Agate Street
Moss Street
Villard Street
Orchard Street
Walnut Street
Columbia Street
Fairmount Boulevard
Skyline Boulevard
Kona Street
Sunset Drive
Fircrest Drive
Birch Lane
Bardell Avenue
Park
Leaflet | © OpenStreetMap, ODbL

Getting started with R

Starting R

Installation

- Install `R`.
- Install `RStudio`.
- Install `Quarto`.
- **Optional/Overkill:** `Git`
  - ↳ Create an account on `GitHub`
  - ↳ Register for a student/educator discount.
  - ↳ For installation guidance and troubleshooting, check out Jenny Bryan's website.
- **Note:** Many UO labs have R installed and ready (helpful in a pinch).

Starting R

Resources

Free(-ish)

- Time
- Google (which inevitably leads to StackOverflow)
- ChatGPT/Codex, Claude, Copilot, and other AI assistants
- Data services at the UO library
- Your classmates
- Your GE and me
- R resources here and here
- swirl and learnr

Starting R

Some R basics

You will dive deeper into R in lab, but here are six big points about R:

1. Everything is an **object**. `foo`
2. Every object has a **name** and **value**. `foo = 2`
3. You use **functions** on these objects. `mean(foo)`
4. Functions come in **libraries** (**packages**) `library(dplyr)`
5. R will try to **help** you. `?dplyr`
6. R has its **quirks**. `NA; error; warning`

Next: Metrics review(s)